

YUNBO LONG

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EDUCATION

University of Cambridge, Cambridge, UK

Oct 2023 – Present

PhD in Engineering

- **Generative Models:** Continuous diffusion for conditional language models (**Latent-DiffRWKV**), Post-training framework for the tabular language models (**TabGRAA**), a geodesic one-step generation for mixed-type data (**ExpoTab**).
- **Autonomous Multi-Agent Systems:** Built **Veritrace**, achieving top 1 in the Deep research Benchmark for open-source MAS (Hugging face). **Helicase**, a multi-agent uncertainty framework for evolving knowledge graphs with consensus-driven convergence. **AI-Supervisor**, an autonomous research agent for self-directed scientific discovery.
- **Agentic Game theory:** Architected Bayesian emotional intelligence (**EmoDebt**), and multi-agent negotiation (**EQ-Negotiator**), agentic evolving systems with emotional personas (**EvoEmo**), MAS for edge-deployable Negotiation systems (**EmoMAS**), Judge Policy Optimization for Emotional Negotiation Skill Distillation from LLMs to SLMs (**EmoDistill**).
- **Award:** Queen's College Scholarship & Department of Engineering Conference Funding

University of Cambridge, Cambridge, UK

Sep 2022 – Aug 2023

MPhil in Engineering

- Dissertation: Clustered federated learning for heterogeneous data. Supervised by SCAIL Lab & Flower Lab.

University of Birmingham, Birmingham, UK

Sep 2019 – Jun 2022

BSc in Engineering — First-Class Honours

- **Frank Wallace Scholarship:** 1st in academic performance (Yr 2–3). Robotics Lab Research Assistant.

INDUSTRY EXPERIENCE

Perceptual Intelligence Lab, Univ. of Bath, Bath, UK

Oct 2025 – Mar 2026

Research Intern — Supervised by Prof. Wenbin Li

- Built on-device LLM orchestrator (0.5B–3B) dynamically coordinating local tool calls — depth estimation, object detection, semantic reasoning, path planning — for legged robot navigation via agent distillation, replacing GPT-4o at 10× lower latency.

Shanghai Academy of AI for Science (SAIS)

Sep 2025 – Dec 2025

AI Research Intern

- Developed diffusion latent language modeling methods by aligning VAE latents with token representations via MMDiT cross-attention and training DDPMs over latent variable z . Also studied latent-state alignment for RWKV and linear-attention models, learning α -based state transitions to model hidden-state changes with diffusion.

DIAL Lab, University of Cambridge, Cambridge, UK

Jun 2023 – Sep 2023

ML Engineer Intern

- Built visual anomaly detection system with unsupervised learning + Anomalib, reducing annotated data needs for industrial inspection.

DataGrand (\$160M+ AI Startup), Shanghai, China

Jun 2022 – Sep 2022

NLP Intern

- Developed OCR text recognition algorithms for enterprise document processing pipelines serving Fortune 500 clients.

SELECTED PUBLICATIONS

- Zhao H*, Long Y*, Lucas B. & Brintrup A. *VeriTrace: Evolving Mental Models for Deep Research Agents*. Under Review
- Long Y., Zhao H., et al. *EmoDistill: Offline Emotion Skill Distillation for Language Model Agents in Adversarial Negotiation*. Under Review
- Long Y. *AI-Supervisor: Autonomous AI Research via a Persistent Research World Model*. ArXiv 2026
- Hao G., Shang Y., Long Y., et al. *Self-Policy Distillation via Capability-Selective Subspace Projection*. Under Review
- Long Y. *, Afonja T. *, Fritz M. & Brintrup A. *Self-Improving Tabular Language Models via Iterative Reward-Guided Post-Training*. Under Review
- Long Y., Liu Y., Xu L. *EmoMAS: Emotion-Aware Multi-Agent System for Edge-Deployable Negotiation*. **ACL Main 2026 (LLM Agent Track)**
- Long Y. *, Zhang J. *, Chen X. & Brintrup A. *Topological Federated Clustering via Gravitational Potential Fields*. **AAAI Main 2026**
- Long Y., Xu L., Lucas B. et al. *EvoEmo: Evolved Emotional Policies for Multi-Turn Negotiation*. **AAMAS EA 2026**
- Long Y., Liu Y., Xu L. & Brintrup A. *EmoDebt: Bayesian-Optimized EQ for Debt Recovery*. **AAMAS Main 2026 (Oral)**
- Long Y., Xu L. & Brintrup A. *Efficient and Privacy-Preserved Link Prediction via Condensed Graphs*. **ESWA 2026**
- Long Y., Liu Y., Lucas B. et al. *EQ-Negotiator: Emotional Personas for Credit Negotiation*. **NeurIPS@PersonaLLM 2025**
- Long Y., Xu L. & Brintrup A. *Evaluating Inter-Column Logical Relationships in Synthetic Tabular Data*. **ICLR 2025@SynData**

TECHNICAL SKILLS

ML/AI: LLM fine-tuning (SFT, RLHF, DPO), Agent systems (ReAct, tool-calling, multi-agent), Knowledge distillation, Diffusion models, RL, Federated learning, Embodied AI, OCR/NLP

Engineering: PyTorch, HuggingFace (Transformers, PEFT, TRL), LangGraph, vLLM, Docker, Linux, W&B, Habitat-sim, Isaac Sim/ROS

Hobbies: Student Formular 1, Avalon